KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) As an application developer, why do you choose to run an application over UDP rather than TCP. (06) [CO1]
   b) Explain the concept of piggybacking in data communication. Discuss how it affects network congestion. (05) [CO2]
   c) Discuss how TCP detects congestion in a network. Describe the congestion control techniques used in TCP to manage and avoid congestion. (12) [CO3]
   d) An organization's network is connected to the internet through a 25 Mbps access link. Users request objects of average size 800 Kbits at 30 requests/second, with average internet delay 1.5 seconds. The organization's internal LAN has a speed of 1 Gbps. To reduce response time, the organization considers: (12) [CO4]
      Option 1: Upgrade the access link to 100 Mbps.
      Option 2: Install a local web cache that serves 40% of requests.
      Compare the average end-to-end delay for both options and recommend which provides better performance and is more cost effective.

2. a) Explain how a TCP connection is established, detailing the role of SYN and ACK flags. Justify why a two-way handshake is not sufficient for reliable communication. (08) [CO2]
   b) Describe why flow control is required in TCP implements flow control between sender and receiver, specifying the conditions that must be met, and illustrate your answer with a suitable example. Additionally, explain the concept of silly window syndrome. (13) [CO3]
   c) In TCP, *EstimatedRTT*, *DevRTT* and *TimeoutInterval* adapt to changing network conditions. Suppose, three successive *SampleRTT* measurements are recorded as 110 ms, 95 ms, and 120 ms. Given: initial $EstimatedRTT = 100\ ms$, initial *DevRTT* = 5 ms, $\alpha = 0.125$, and $\beta = 0.25$ (14) [CO3]
      i) Compute *EstimatedRTT*, *DevRTT* and *TimeoutInterval* after each *SampleRTT*
      ii) Explain why it is not advisable to use the RTT of retransmitted packets when updating *TimeoutInterval*

3. a) Discuss why DNS is used in networking. Illustrate the hierarchy of DNS servers and describe how a DNS query is resolved. (09) [CO2]
   b) Explain the working of the Selective Repeat (SR) protocol. Given a sequence number space of 4 and a sender window size of 3, determine whether the window size is appropriate for SR protocol. Discuss the problems that can occur if the window size exceeds the allowed limit with a timeline diagram. (13) [CO3]
   c) Are IP address alone sufficient for communication between applications? Suppose, you have a web server that hosts three web applications on ports 3000, 5000, and 8000, but for some reasons users can't access them via standard HTTP/HTTPS requests using the domain name. Identify the underlying issue and recommend an appropriate solution to enable access via standards ports. Illustrate your approach with a request-routing diagram. (13) [CO4]

4. a) While searching for new pair of shoes, you visited multiple e-commerce sites and may or may not have completed a purchase. Later, when browsing a social media platform, you noticed advertisements for the same types of shoes you had viewed earlier. Explain the underlying mechanism that enables such personalized advertisements across different websites. (10) [CO3]
   b) Explain the rules for generating acknowledgements in TCP and discuss the fast retransmit mechanism. Illustrate your answer with suitable examples. (14) [CO3]
   c) Describe how man-in-the-middle attack hinders the authentication process (ap5.0) that uses a nonce and public key cryptography. (11) [CO3]

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SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the necessity of a trunk port in VLAN. Compare "Router-on-a-Stick" and "Layer 3 Switch (SVI)" based on following criteria: (10) [CO1]
   i) Scalability
   ii) Performance/Latency
   iii) Implementation Complexity
   iv) Cost

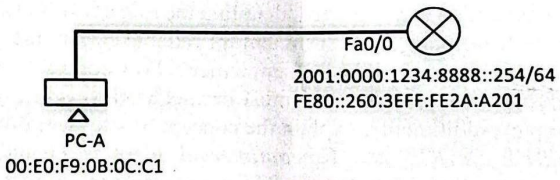
   b) Explain HOL Blocking. Analyze the major switching techniques with their benefits and drawbacks. (10) [CO1]

   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14 bytes header and has an MTU of 1024 bytes; the second network uses 8 bytes header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP headers are 20 bytes. (15) [CO3]

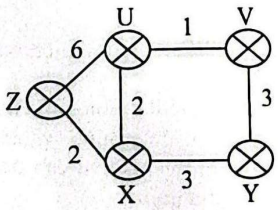
6. a) "IPv6 is faster and more secure than IPv4" – justify the statement based on IP datagram header. (05) [CO2]

   b) What are the differences between IP based forwarding and generalized forwarding? Explain simple packet handling rules for generalized forwarding based on OpenFlow protocol. (10) [CO3]

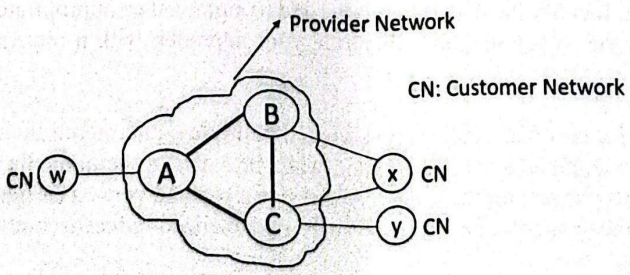
   c) What is link-local address in IPv6? Describe the process of Router Solicitation Message showing IPv6 header and ICMPv6 header values to get an IP using Stateless Auto Configuration mode. Calculate the IP of PC-A during this process shown in following figure. (20) [CO4]



7. a) Consider the network shown in following figure and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12) [CO3]

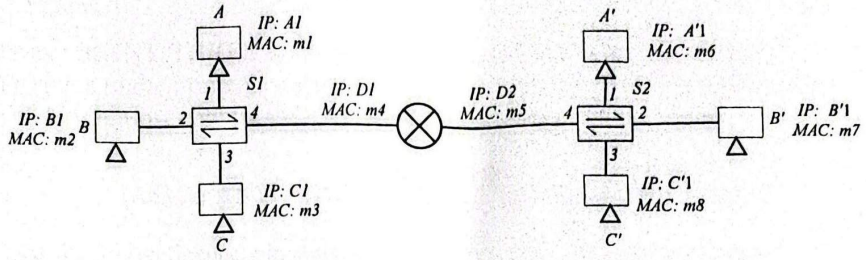


   b) What is Hot Potato routing? Describe the policy-based path selection for the following scenario. (10) [CO2]

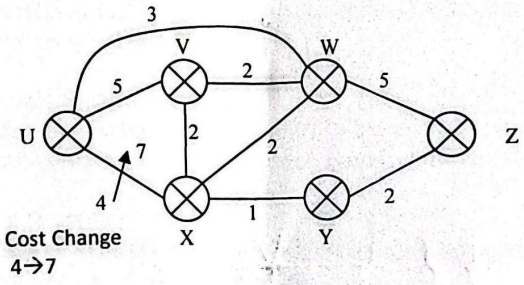


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c) You have to look carefully the following figure. Computer A wants to send a frame to Computer C'. Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? Initially switch table was empty. Finally, write down the self-learning algorithm of switch. (13) [CO2]



8. a) Explain Layer 2.5 protocol-based packet forwarding operation using an example. (10) [CO3]
   b) Emulate the Software Define Network (SDN) controller's interaction with local switches to modify the link-state changes occurred in the following network. Apply Dijkstra's link-state algorithm in the SDN controller to calculate the updated routing table for 'U'. (17) [CO4]



c) State how packet filtering firewall works. (08) [CO1]